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An 8-Week Neuromuscular Training Program After Concussion Reduces 1-Year Subsequent Injury Risk

A Randomized Clinical Trial

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Background: Existing data suggest that after concussion, athletes experience an increased risk of subsequent injury. Exploring methods that may reduce injury risk after successful postconcussion return to play may lead to new treatment approaches.

Purpose: To examine the efficacy of a neuromuscular training (NMT) intervention on acute sports-related time-loss injury over the subsequent year relative to standard of care.

Study Design: Randomized clinical trial; Level of evidence, 1.

Methods: A total of 27 youth athletes were assessed initially postconcussion (median, 7 days postconcussion; interquartile range [IQR], 5-10) and after return-to-play clearance (median, 40 days postconcussion; IQR, 15-52). After return-to-play clearance, they were randomly assigned to NMT intervention (n = 11; mean ± SD age, 14.7 ± 1.7 years; 36% female) or standard of care (n = 16; mean ± SD age, 15.3 ± 1.8 years; 44% female). The intervention (duration, 8 weeks; frequency, 2 times per week) included guided strength exercises with landing stabilization focus. Standard of care received no recommendations. For the subsequent year, athletes prospectively completed a monthly log of sports-related injuries and organized sports competitions.

Results: During the first year after postconcussion return-to-play clearance, sports-related time-loss injuries were more common among standard of care relative to NMT intervention (75% [95% CI, 48%-93%] vs 36% [95% CI, 11%-69%]). After adjusting for age and sex, the hazard of subsequent injury in the standard-of-care group was 3.56 times (95% CI, 1.11-11.49; P = .0334) that of the NMT intervention group. Sports participation was similar between NMT intervention and standard of care during the year-long monitoring period (hours of organized sports per month; median, 12 [IQR, 2.6-32.1] vs 15.6 [IQR, 3.5-105.9]; P = .55). The age- and sex-adjusted incidence of injuries was 10.2 per 1000 competitive exposures (95% CI, 3.7-28.4) in the standard-of-care group as opposed to 3.4 per 1000 (95% CI, 0.9-13.4) in the NMT intervention group. After adjusting for age and sex, incidence of injuries was higher for standard of care vs NMT intervention (rate ratio, 2.96 [95% CI, 0.89-9.85]; P = .076).

Conclusion: Although preliminary, our findings suggest that an NMT intervention initiated after return-to-play clearance may significantly reduce sports-related time-loss injuries over the subsequent year.

Registration: NCT03917290 (ClinicalTrials.gov identifier).

Keywords: mild traumatic brain injury; adolescent; injury prevention; rehabilitation; treatment; therapy; exercise

Concussion, a type of brain injury, is defined as a traumatically induced alteration of mental status that may or may not involve loss of consciousness,[2] although many definitions for concussion exist.[31] Given the heterogeneity of clinical presentation and the multidimensional nature of the injury, clinical decision making can be challenging. Evolving clinical practice guidelines over the past 2 decades may have helped to reduce the risk of recurrent concussions.[30] However, other negative consequences after concussion have been identified for athletes and remain challenging to treat. Of particular interest, existing meta-analyses indicate that athletes who return to play (RTP) after concussion have a greater risk of subsequent sports-related injuries than those without a concussion.[33,39] Existing data demonstrate associations between

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concussion and increased odds of acute sports-related time-loss lower extremity injuries, including anterior cruciate ligament tears, ankle sprains, or lower extremity muscle strains.[9,20,34,37]

Evidence suggests that the incidence of sports-related injuries is higher among athletes postconcussion versus those without concussion, but the reasons why this occurs are not fully understood, and approaches to mitigate this risk have not been implemented to date. Persistent neuromuscular control deficits have been documented after RTP clearance postconcussion,[15,16] and such deficits may contribute to an increased subsequent injury risk during sports. Three studies have identified an association between neuromuscular/attentional deficits (dual task) after RTP clearance and subsequent sports-related injury risk in the year after concussion.[12,13,36] Others have proposed that the following may contribute to increased postconcussion injury risk: delayed muscle integration owing to motor cortex dysfunction,[49] perception-action coupling deficits that lead to improper body position or motor execution,[7] or diminished motor task execution abilities.[4] Collectively, it appears that inefficient integration of brain networks may relate to increased injury susceptibility after a sports-related concussion.

One commonality among proposed hypotheses for the increased incidence of sports-related injuries after a concussion is that movement control dysregulation occurs after a concussion and persists for a period that may outlast clinical recovery. This may relate to cognitive and motor factors, as attention, orienting, awareness, or functional adaptability may influence injury risk.[5] Given this, an intervention that targets the restoration of neuromuscular control abilities may be a viable rehabilitation strategy to reduce the risk of future injuries. The majority of current concussion rehabilitation research focuses on aerobic exercise during recovery to reduce symptoms.[25,27,42] Although beneficial, this approach does not consider neuromuscular factors. Currently, no evidence exists examining the effect of neuromuscular training (NMT) on injury risk initiated upon RTP after a concussion. Previous work using NMT interventions among adolescent athletes without concussion suggests that this approach can substantially reduce musculoskeletal injury rates[8] and concussion incidence.[11] Given the particular vulnerability to musculoskeletal injury after a concussion,[33,39] NMT deployed after concussion is a reasonable strategy to reduce the risk of subsequent injuries.

To date, no interventions have been proposed to modulate subsequent injury risk postconcussion. Therefore, the primary purpose of our pilot study was to examine the efficacy of an 8-week NMT intervention on acute sports-related time-loss injuries sustained over the subsequent year as compared with standard of care (no intervention) via a prospective stratified randomized clinical trial. We hypothesized that the hazard of injury in the year after postconcussion RTP would be significantly greater for those randomized to standard of care relative to NMT.

METHODS

Trial Design and Participants

We conducted a block-stratified randomized clinical trial in which we recruited and enrolled participants within 14 days of sustaining a concussion. Participants were recruited from and treated within a single sports medicine clinic (Children's Hospital Colorado) associated with a tertiary care regional hospital between April 23, 2019, and March 6, 2020; the final 1-year follow-up injury log was completed on March 31, 2021. All concussions were diagnosed by board-certified sports medicine physicians, who made all clinical decisions regarding return to sports independent of the study. Concussion definition and management practices were consistent with recommendations provided by the most recent international consensus statement on concussion in sport during the study period.[32]

Our inclusion criteria consisted of enrollment $\leq$14 days postconcussion, participants who were 12 to 18 years of age at the time of enrollment (secondary/high school student-athletes), confirmed concussion diagnosis by a sports medicine physician, and a postconcussion symptom inventory score $\geq$9 to ensure that participants were symptomatic at the time of enrollment. Patients who had recovered by the time of the first assessment were not included so that our sample had a relatively homogeneous profile upon enrollment, to the degree possible. Our exclusion criteria consisted of a concomitant sports-related injury, concussion within 6 months of the current injury, a documented structural brain injury confirmed via neuroimaging, a concussion sustained during a high-velocity mechanism (not sportlike), a second head injury before clearance for the current injury, or intention to retire from organized sports. The study protocol was registered via ClinicalTrials.gov (NCT03917290) and approved by the local institutional review board (18-2680) before commencement. All

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participants and parents/guardians, if participants were aged <18 years, provided informed consent/assent.

Interventions

Athletes were randomized to NMT intervention or standard of care (allocation ratio 1:1) after they received RTP clearance from the treating physician. No study interventions were performed between the initial visit and the RTP clearance visit. Athletes reported injuries and athletic competition volumes monthly over the subsequent year via a REDCap survey (Research Electronic Data Capture; Vanderbilt University Medical Center)[10] (Appendix Figure A1, available in the online version of this article).

The NMT intervention was adapted from existing programs that focus on plyometric, strength, technique, and balance training.[8,35] This approach has been used among youth athletes without concussion to achieve neuromuscular system adaptations and reduce sports injury risk.[8,35] Beyond these training elements, we incorporated a dual-task progression as a single exercise at the end of the training program. Dual tasks have been theorized to assist with concussion rehabilitation[21] and have been efficacious in other brain injury rehabilitation studies.[47] The dual task consisted of motor progressions (standing, walking, hopping, ball catching, side-to-side ball catching) and cognitive progressions (animal naming, digits backward, immediate memory recall) performed simultaneously. These cognitive tasks were selected as progressively more difficult tasks throughout the rehabilitation progression and in line with past work using dual tasks as a postconcussion assessment approach.[17] A full description of the specific exercises performed and the rehabilitation progression is available in our feasibility trial publication.[18]

Those randomized to the intervention group met with a trained member of the study team (including author C.S.) twice per week for 8 weeks to complete NMT. The study team member supervised all exercises; provided feedback and instruction regarding volume, pace, and form; and documented compliance. Most participants (82%) in the NMT intervention group received in-person instruction. However, given the cessation of in-person clinical research activities because of the COVID-19 pandemic in March 2020, 2 intervention participants underwent NMT via telehealth; specifically, the participant performed exercises at home while a study team member (C.S.) provided live, real-time feedback using a video feed. The intervention was not altered except for the real-time feedback and instruction provided to participants in this telehealth environment, rather than in person; otherwise, the progressions remained the same. If necessary, comparable household objects were substituted for equipment as needed for study visits conducted via telehealth. Those randomized to standard of care were instructed to comply with their physician recommendations regarding RTP and received no specific instructions from the study team. Physician recommendations were consistent with the RTP strategy outlined in the most recent international consensus statement on concussion in sport during the study period.[32]

Primary Outcome Variable

After RTP clearance, athletes prospectively completed a monthly online survey for the subsequent 12 months. Given that athletes were participating in organized sports through a variety of settings, this questionnaire-based approach allowed for prospective and continuous monitoring of injury exposure and incidence throughout the year, improving on past approaches that relied on a 1-year recall of postconcussion sports-related injuries.[12,13]

Each month, participants were asked to complete an online form (via REDCap[10]) in which they reported any new time-loss injuries that they had incurred during sports (ie, missed ≥1 practice or game because of the injury) and if they participated in organized sports during the month (ie, with a coach and structured practice). If they selected yes to either item, we obtained further details. For injuries sustained, the following details were provided: date of injury, diagnosis, medical personnel providing diagnosis, sport played during the time of injury, time missed in organized sports because of the injury, and a narrative description of what happened to confirm details provided. For our primary and secondary analyses, we included only sports-related injuries that resulted in time lost from competition. If participants noted that they had participated in organized sports during the month, they reported the number of practices, number of hours practicing, number of games, and number of hours spent playing games (broken down into each sport if they indicated multiple sports in a given month).

Our primary outcome variable was whether a sports-related time-loss injury occurred (yes or no) during the 1-year monitoring period and, if a subsequent injury occurred, the time from postconcussion RTP clearance until the injury. Secondary outcome variables included the incidence of sports-related time-loss lower extremity musculoskeletal injuries as well as the total number of injuries sustained over the year of monitoring and the injury severity (ie, number of days lost attributed to injury). Total injury incidence was estimated as the number injuries per 1000 competitive exposures. The total number of competitive exposures per athlete was estimated as the sum of all practices and games in which the athlete reported participating when completing the monthly form during the year after postconcussion RTP. As a secondary purpose, we examined the efficacy of the NMT intervention to reduce the risk of acute sports-related time-loss lower extremity musculoskeletal injury, given that most of the existing work in this area has focused on such injuries.

Primary Endpoint

In this exploratory investigation, our primary endpoint was at 1 year after clearance to return to sports. Participants lost to follow-up before reaching this primary

endpoint were included as censored data in the time-to-event analysis.

Participant Characteristics

To ensure that the randomization scheme was effective, we compared the distribution of potential confounding variables in the 2 groups: demographic characteristics (age, body size, sex, race, ethnicity, and whether athletes had begun physical activity since their concussion), injury history before the concussion for which they were being treated at enrollment (diagnosed sports-related concussions and sports-related time-loss musculoskeletal injuries), relevant medical history (diagnosed preconcussion headache disorder, migraine, attention-deficit disorder or attention-deficit/hyperactivity disorder, anxiety, or depression), and initial concussion severity (loss of consciousness or amnesia at the time of injury, time to symptom resolution and time to RTP, initial symptom severity via the Post-concussion Symptom Inventory, and 5P persistent symptom risk score).[19,22,41,51] The 5P persistent symptom risk score—a test that uses 9 variables to predict the risk of developing persistent symptoms after a concussion—was included for its clinical utility and strength as a prognostic approach to expected concussion symptom recovery among adolescents.[19,51]

Sample Size and Randomization

Before study commencement, we calculated our required sample size using an estimate from an injury prevention program among youth athletes conducted over a 1-month period[38]; this coincided with the feasibility and functional measurement acquisition arm of our overall study, which examined pre- vs postintervention performance, and has been published.[18] The variable of interest was vertical drop jump height, used as a surrogate for future injury risk based on the absence of previous data allowing for power calculation based on injury risk alone. As such, we anticipated that improvements in this variable would relate to a reduction in injury risk after a concussion. In this previous study, the authors reported a pre- vs postintervention change on drop jump performance that was greater for the intervention group than the control group (Cohen d = 1.05). Given an alpha of .05 and power of 0.80, these results suggest that enrolling 32 participants would be necessary to achieve our primary aim. With an estimated 80% retention rate, we registered our study to include a target of 40 participants enrolled, with the expectation that we would have 32 in the final analysis. However, with the cessation of research activities in March 2020 attributed to the COVID-19 pandemic, we were unable to continue enrollment beyond the first 32 participants. We therefore conducted our analysis once we had enrolled 32 participants in the study who were initially tested before March 6, 2020, and who continued completing monthly injury questionnaires over the subsequent year.

We used a block-stratified randomization procedure, with a block size of 4 and stratification variables of sex (female vs male) and age (12-15 vs 16-18 years). Given the nature of the intervention, no blinding procedures were used during data collection or intervention administration, nor was this possible or meaningful. The lead author (D.R.H.) generated the random allocation sequence, and investigative team members enrolled participants and assigned them to each group independent of the lead author. All statistical analyses were performed after the 1-year follow-up period was complete, in line with the preplanned analysis strategy by a member of the study team (P.M.C.), who was blinded to group assignment labels to ensure bias reduction to the degree possible.

Statistical Analysis

Descriptive statistics were used to summarize participant and clinical characteristics in the intervention and standard-of-care groups. Based on the pilot nature of the data collection, a per-protocol analysis was used to compare subsequent injuries in the 2 groups. We also performed an intention-to-treat analysis, with groups determined per randomization assignment. Cox proportional hazards regression analysis was used to test for differences in the hazard of any injury between the intervention group and the standard-of-care group. It was possible for athletes to experience multiple injuries during the follow-up period. We tested for group differences in the number of all injuries as well as the occurrence of lower extremity injuries specifically. Negative binomial models were used to test for group differences in the number of all injuries as well as the number of lower extremity injuries during the follow-up period. The total number of competitive exposures and the sum of the total number of competitions and practices were used as an offset term to calculate an estimate of the incidence of injuries per 1000 competitive exposures. Owing to the stratified nature of the randomization assignment, all analyses were adjusted for the age (12-15 and 16-18 years) and sex randomization strata. The data analysis was generated using SAS software (Version 9.4; SAS Institute Inc).

RESULTS

We assessed 55 patients for eligibility and interest in the study during the enrollment period, and 32 enrolled. Among those who enrolled, 4 did not return for the RTP clearance visit and thus were not randomized (lost to follow-up), leaving 13 participants allocated to the NMT intervention and 15 to standard of care (Figure 1). One participant randomized to standard of care was lost to follow-up (ie, did not complete any monthly injury logs). Two participants initially randomized to the NMT intervention elected to not participate in the intervention but continued with the longitudinal tracking via monthly injury logs and thus were placed into the standard-of-care group (per-protocol analysis) (Figure 1). In total, we included 11 NMT intervention participants and 16 standard-of-care participants in our

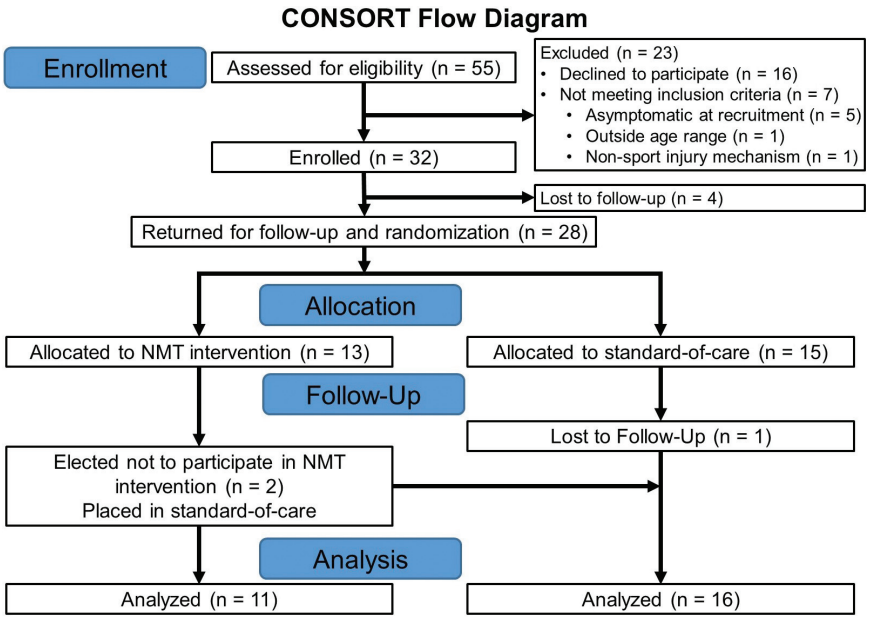

Figure 1. CONSORT (Consolidated Standards of Reporting Trials) diagram. NMT, neuromuscular training.

final analyses. Thus, a total of 27 youth athletes were assessed initially postconcussion (median, 7 days postconcussion; interquartile range [IQR], 5-10) and after return-to-play clearance (median, 40 days postconcussion; IQR, 15-52). No adverse events during the intervention sessions or unintended effects caused by study involvement were observed. The NMT intervention, on average, lasted 59 days (SD, 13); each training session required a mean 18 minutes (SD, 5) to complete; and 91% (n = 10/11) of participants completed ≥75% of the intervention sessions (a mean 88% of the 16 intervention sessions were completed).

The groups were similar in many demographic characteristics, such as age, sex, race, and body mass index. The groups differed with respect to participation in physical activity before initial evaluation, as well as having a history of multiple sports-related concussions; specifically, the NMT intervention group had a greater proportion of patients already participating in physical activity as well as a greater proportion with a history of multiple concussions (Table 1). Athletes were recruited from a diverse array of sports played at the time of the initial concussion (Table 2). In the intervention group, the initial concussion occurred most frequently during participation in football, whereas in the standard-of-care group, the initial concussion occurred most frequently during participation in soccer. The timing of the intervention group at initial visit (median, 9 days postconcussion; range, 3-13) and RTP clearance/randomization (median, 22 days postconcussion; range, 14-93) was similar to that in the standard-of-care group (median, 6 [range, 3-11] and 42.5 [11-140], respectively).

The duration of the follow-up injury monitoring period (days from postconcussion RTP clearance until final follow-up monthly log completion) was similar between intervention (median, 366 days; range, 336-382) and standard of care (median, 369 days; range, 363-387) groups. Based on the Wilcoxon rank-sum test, compliance (ie, the number of complete months of activity/injury reporting) was also similar in the 2 groups ($Z = -0.0365$; $P = .9711$). The median number of months with complete injury/activity data was 12 (range, 11-12) in both groups.

After RTP clearance, injuries were more common after standard of care (75%; 95% CI, 48%-93%) than intervention (36%; 95% CI, 11%-69%). After adjusting for the randomization strata (age and sex), the hazard of any subsequent injury in the standard-of-care group was 3.56 times (95% CI, 1.11-11.49; $P = .03$) that in the NMT intervention group (Figure 2). Ankle sprains were the most common injury between groups during the follow-up period (Table 3). Within the first 90 days of postconcussion RTP, no intervention group participants sustained an injury, as opposed to 50% (n = 8/16; 95% CI, 25%-75%) in standard of care.

It was possible for participants to sustain multiple injuries during the 1-year follow-up period. Therefore, we tested for differences in the total number of injury events between the groups. Given the small differences in sports participation exposure history between groups (Table 4; average h/mo of games and practices $p = 0.55$), we estimated the total number of injuries relative to the total number of competitive exposures. In the standard-of-care group, the incidence of injuries was 10.2 per 1000 competitive exposures (95% CI, 3.7-28.4) as opposed to 3.4 (95% CI, 0.9-13.4) in the NMT intervention group. After adjusting for age and sex, incidence of injuries was also higher in standard of care group relative to the NMT intervention group (rate ratio, 2.96 [95% CI, 0.89-9.85]; $P = .0762$]. In the NMT intervention group, only 1 participant sustained >1 injury in the year after postconcussion RTP, as opposed to 5 in the standard of care group.

During the intervention period (ie, the first 2 months of injury monitoring after RTP/randomization), the NMT

TABLE 1
Participant Characteristics: NMT Intervention and Standard of Care[a]

Variable	NMT Intervention (n = 11)		Standard of Care (n = 16)	
Demographics/clinical characteristics				
Mean ± SD[b]				
Age at initial visit, y	14.7	1.7	15.3	1.8
Body mass index	23.3	6.3	21.8	4.7
No. (%)[c]				
Female sex	4	36	7	44
White	6	55	12	75
Hispanic ethnicity	3	27	3	19
Physical activity before initial evaluation	5	45	3	19
Injury/concussion history				
History, No. (%)[c]				
Sports-related concussion	5	45	11	69
Multiple sports-related concussions	5	45	3	19
Sports-related musculoskeletal injury	10	91	12	75
Preconcussion relevant medical history				
Diagnosis, No. (%)[c]				
Headache disorder	3	27	4	25
Migraine	3	27	1	6
ADD or ADHD	0	0	1	6
Anxiety	2	18	2	13
Depression	2	18	3	19
Concussion severity				
No. (%)[c]				
Loss of consciousness	3	27	2	13
Amnesia	6	55	4	25
Median (IQR)[d]				
Time from concussion to symptom resolution	17	12-43	19	10.5-46.0
Time from concussion to RTP clearance	22	15-49	35	15-60
Initial symptom severity	46	24-74	41	29-73
5P score at the initial study visit	7	7-10	8	7-10

[a]ADD, attention-deficit disorder; ADHD, attention-deficit/hyperactivity disorder; IQR, interquartile range; NMT, neuromuscular training; RTP, return to play.
[b]Student *t* test.
[c]Fisher exact test.
[d]Wilcoxon rank-sum test.

TABLE 2
Sports Participation When the Concussion Occurred[a]

	NMT Intervention (n = 11)		Standard of Care (n = 16)	
Sport	No.	%	No.	%
Soccer	2	18	7	44
Basketball	1	9	2	13
Football	4	36	3	19
Volleyball	0	0	1	6
Cheerleading	0	0	2	13
Wrestling	1	9	0	0
Ultimate frisbee	1	9	0	0
Baseball	2	18	0	0
Karate	0	0	1	6

[a]NMT, neuromuscular training.

intervention and standard-of-care groups reported similar data for the following (median [range], respectively): number of practices (20 [3-40] vs 22 [8-33]), number of games played (5 [0-9] vs 6 [0-10]), and average hours per month playing organized sports (26.5 [7-59.5] vs 25.5 [14.5-44]). In addition, we did not observe any differences between groups in exposure over the final 10 months of post-RTP monitoring (number of practices, $P = .53$; games played, $P = .46$; hours per month playing organized sports, $P = .42$).

Finally, we tested for differences in the incidence of a time-loss lower extremity musculoskeletal injury between groups. The age- and sex-adjusted incidence was 6.2 per 1000 competitive exposures (95% CI, 1.7-22.3) in the standard-of-care group as opposed to 2.4 per 1000 (95% CI, 0.5-12.6) in the NMT intervention group. Please insert results of statistical significant testing for this univariable association. After adjusting for age and sex, the incidence of lower extremity injuries was higher for standard of care relative to NMT (rate ratio, 2.55 [95% CI, 0.54-12.16]; $P = .2394$). Appendix Table A1 (available online) provides a summary of the unadjusted and adjusted hazard of injury, incidence of injury, and incidence of lower extremity injuries.

TABLE 3
Subsequent Injuries Reported by Participants Over the 1-Year Postconcussion RTP Monitoring Period[a]

	NMT Intervention (n = 4)		Standard of Care (n = 12)	
	No.	%	No.	%
Time loss				
<1 wk	2	50	7	58
1-2 wk	2	50	3	25
1-2 mo	0	0	2	17
Sport				
Basketball	0	0	1	8
Bicycle	0	0	2	17
Football	0	0	2	17
Lacrosse	1	25	0	0
Martial arts	0	0	1	8
Soccer	1	25	4	33
Surfing	1	25	0	0
Tennis	1	25	0	0
Volleyball	0	0	1	8
Wrestling	0	0	1	8
Injury type				
Concussion	0	0	1	8
Dislocation: shoulder	1	25	0	0
Fracture				
Hand	0	0	1	8
Rib	0	0	1	8
Sprain				
ACL	0	0	1	8
Ankle	1	25	4	33
Knee	1	25	0	0
Strain				
Achilles tendon	0	0	1	8
Hip flexor	1	25	0	0
Rib	0	0	1	8
Shoulder	0	0	1	8
Thigh	0	0	1	8

[a]ACL, anterior cruciate ligament; NMT, neuromuscular training; RTP, return to play.

On the basis of the pilot nature of the study, we presented the per-protocol analysis to illustrate the potential efficacy of the intervention among athletes who actively participated in the intervention. Appendix Table A2 (available online) provides the results of the intention-to-treat analysis, with groups determined by randomization assignment. Overall, the conclusions drawn from the intention-to-treat analysis are consistent with the per-protocol analysis. The hazard of an injury in the year after concussion was significantly higher for standard of care relative to NMT; furthermore, the incidence of all injuries, as well as the incidence of lower extremity injuries, was higher for standard of care relative to NMT.

DISCUSSION

Data from our investigation indicate that an NMT intervention initiated upon RTP from a sports-related concussion, administered 2 times per week for 8 weeks, was protective against sustaining a sports-related time-loss injury over the subsequent year relative to standard of care. However, we did not identify significant differences in injury incidence between groups for time-loss lower extremity musculoskeletal injuries. The effects observed suggest that this is a feasible and promising approach to reducing the well-documented burden of postconcussion injuries.[33,39]

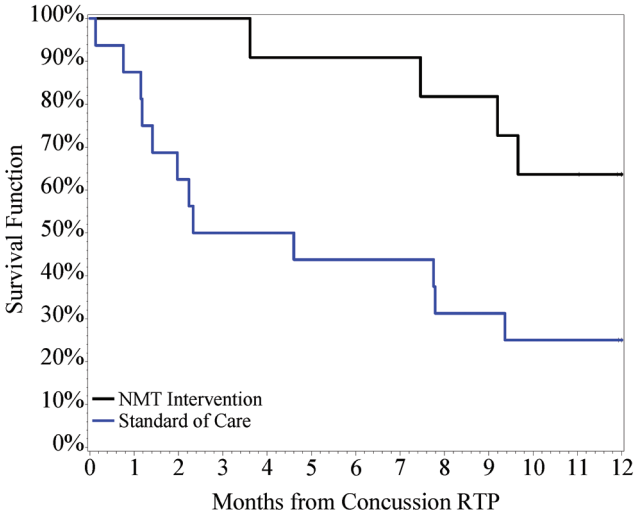

Figure 2. Injury-free survival between standard of care and NMT intervention. NMT, neuromuscular training; RTP, return to play.

The rehabilitation approach that we used among athletes who were recently concussed was like that in studies among female high school athletes without concussion who played volleyball, soccer, or basketball. This previous work indicated that NMT participation throughout a competitive season resulted in a lower incidence of knee/ankle injury relative to a sham protocol.[8] As athletes are susceptible to sports-related injuries after concussion,[1,3,6,23,29,50] the relative effect of NMT in athletes with recent concussion as compared with standard of care may be even greater than previous observations among youth athletes who perform NMT throughout the season.[43] Indeed, in our study we observed a protective effect of NMT intervention against sustaining a sports-related time-loss injury in the year after postconcussion RTP. Although the incidence of postconcussion lower extremity musculoskeletal injuries was higher among the standard-of-care group relative to the NMT intervention group, we did not observe a statistically significant effect.

Based on the well-established model of injury prevention[48] in the context of postconcussion subsequent injuries, step 1 (establish the extent of the problem) has been reported through recent systematic reviews.[33,39] Although several theories[7,15,49] and experimental data[12,13,36] exist related to step 2 (mechanisms responsible for an increased postconcussion injury risk), it is reasonable to begin work examining preventive measures (step 3) that are easily accessed and deployed by a range of health care professionals. Even though we

TABLE 4
Sports Participation Reported Throughout the 1-Year Postconcussion RTP Monitoring Period[a]

	NMT Intervention		Standard of Care	
Variable	Median	Range	Median	Range
No. of practices	72	8-165	68	15-215
No. of games	8	1-53	11	1-51
Games and practices, average h/mo	12	2.6-32.1	15.6	3.5-105.9

[a]NMT, neuromuscular training; RTP, return to play.

identified an increased hazard of injury for the standard-of-care group relative to the NMT intervention group over the year-long monitoring period, time-to-event data in Figure 2 suggest that the most beneficial effect against postconcussion subsequent injuries was during the intervention (initial 60-70 days after postconcussion RTP). As with previous work examining the benefits of NMT among athletes, there may be a dose-response relationship where higher volumes of NMT program participation may have an effect on the relative benefit for reducing sports-related injury risk.[45] In addition, athletes may have been particularly susceptible to subsequent injuries during the initial postconcussion RTP period. For instance, they may have taken substantial time away from sports with subsequent deconditioning effects, or physiological healing of the brain had not yet fully occurred upon RTP clearance.[24] Regardless, our study was not designed to parse out these aspects. The similarity in postconcussion RTP time and athlete exposure volume between the groups during the first 2 months of the monitoring period (ie, when the intervention was occurring) suggests, however, that the effect was not because of exposure differences. Thus, future studies should examine the timing of intervention onset after a concussion, as well as the intervention timing (duration, time of initiation) as individuals recover and reintegrate into sports.

It was challenging to understand unmeasured effects in our study, as many of the participants completed their year-long injury-monitoring period under unprecedented conditions that arose in 2020 because of the COVID-19 pandemic. Although our study was not designed to address effects related to a global pandemic with ensuing stay-at-home orders and restriction of organized youth sports, this experience may have caused unintended or unmeasured influences on the data that we acquired.

Previous meta-analyses indicate that preventive NMT interventions can reduce the incidence of sports-related musculoskeletal injuries, such as anterior cruciate ligament ruptures.[44,46] These benefits occur when training programs incorporate plyometrics, strengthening, and proximal control exercises.[46] After concussion, not only is there a documented increased risk of future injury, but athletes who sustain a concussion and are tested after medical clearance to return to sports have demonstrated greater knee valgus landing characteristics during a double-limb jump-landing task.[28] Even though we did not measure jump-landing performance in our study, the exercises performed may have helped to address the landing/stability deficits that may exist after a concussion. Thus, among this population of athletes who are susceptible to secondary injuries, NMT interventions may represent a viable intervention approach to pursue.

Incorporating NMT interventions after concussion is a reasonable approach, given the low risk of such exercises, the relatively low time burden (40 min/wk for 8 weeks), and the potential benefits. However, these findings should be viewed as preliminary, given the small sample size, single site, and adolescent age range. The feasibility of our approach (most participants attended >75% of the planned training sessions) was higher than previous NMT studies, in which the average compliance rate was 45%.[44] In contrast, adolescents with concussion seemingly adhere poorly to physician recommendations[40] or aerobic exercise interventions.[14] Thus, incorporating NMT elements into RTP protocols may augment existing rehabilitation approaches, which have largely focused on aerobic exercise.[26,42]

Limitations

Our study contained limitations that should be considered when interpreting the findings. We conducted a per-protocol analysis given the small sample size, and as such, results should be considered preliminary. Injuries were self-reported, although the prospective nature of completing monthly injury logs may have mitigated this limitation to some degree. Injury risk prediction can be influenced by many factors, and given the variety of sports played and injuries sustained by participants, specific elements should be accounted for in future investigations, such as sport, body location, and type of injury. We also did not account for contact vs noncontact injuries or any injuries not related to sports, which may have occurred during the monitoring period (1-year follow-up). Participants who completed this study were recruited from a specialty care sports medicine center and may represent a different sample than athletes who seek medical care from different points of entry into the health care system. Furthermore, the feasibility and adherence within our study may not be generalizable to other settings. Finally, our study was not designed to address the restriction of organized youth sports during 2020 that resulted from the COVID-19 pandemic. As a result, we had to alter our protocol wherein 2 participants had to complete the NMT intervention via telehealth, and our sample size was

smaller than what we anticipated when registering our study before commencement. Although athletic competition volumes were similar between groups, organized sports formats may have been adapted to meet social distancing guidelines and thus affected exposure to subsequent injury.

CONCLUSION

After athletes were cleared to return to sports after a concussion, an 8-week NMT intervention demonstrated a significant protective effect in reducing sports-related time-loss injury over the subsequent year. Despite the study limitations, these findings provide initial promising evidence related to an approach for clinicians to consider when developing RTP and rehabilitation programs for athletes who sustain a concussion.

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DATA-SHARING STATEMENT

Deidentified individual-patient data and other study-related documents may be shared upon request from the corresponding author.

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